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ORIGINAL RESEARCH

Prioritizing Risk Factors for Nosocomial COVID-19 Infection in a Taiwanese Hospital: An Integrated DEMATEL–ANP–IPA Approach

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Objective: During the COVID-19 pandemic, the rapid influx of patients with undiagnosed or confirmed infection increased the risk of healthcare-associated transmission and nosocomial infection. Identifying and prioritizing risk factors is essential for strengthening infection prevention and supporting hospital risk management and resource allocation. This exploratory study aimed to prioritize factors associated with nosocomial COVID-19 infection from an expert-based perspective.

Methods: An integrated multicriteria decision-making (MCDM) framework combining the decision-making trial and evaluation laboratory (DEMATEL), analytic network process (ANP), and importance-performance analysis (IPA) was used to examine the relative importance, interrelationships, and intervention priorities of risk factors. Eight experts from five hospitals in Taiwan participated in the assessment. One hospital in Taipei City was selected as the case hospital for exploratory evaluation and prioritization.

Results: "Lack of hospital protective equipment" and "Superspreading event" were identified as the most influential attributes, while "Patients with high viral load" and "Close contact" were the key risk factors associated with nosocomial COVID-19 infection. The integrated MCDM analysis also revealed interrelationships among risk factors and identified priorities of risk factors of infection-control intervention for the case hospital.

Conclusion: The case hospital should prioritize adequate hospital facilities and infection risks associated with patients with chronic diseases when developing nosocomial infection prevention strategies. The proposed framework may support hospital risk assessment, infection-control planning, and resource allocation during infectious disease outbreaks. Given the exploratory and single-hospital design, the findings should be validated in other hospital settings.

Keywords: COVID-19, nosocomial infection, infection control, multi-criteria decision-making, MCDM

Introduction

COVID-19 (coronavirus disease 2019) is a global pandemic that quickly spread to many countries worldwide in early 2020, gradually evolving into a global epidemic. Symptoms of COVID-19 are similar to those of SARS, which occurred in 2002, and the common cold or influenza. This similarity and the emergence of novel viruses may lead to misdiagnosis in patients with other diseases.[1] Hospitals admit many confirmed COVID-19 patients without real-time screening[2] or adequate infection control prevention strategies. This can result in spreading to uninfected patients, leading to nosocomial infections and the collapse of the medical system. This poses a major challenge to the medical systems of countries worldwide. The medical community has recognized the dangers and hazards of COVID-19 for hospital nosocomial infections and government agencies.[3] In the face of limited medical resources and the rapid influx of patients with COVID-19, hospitals need to identify and prioritize factors associated with nosocomial infection to support timely infection prevention, risk management, and resource allocation. However, hospital nosocomial infection involves multiple interrelated clinical, environmental, and organizational factors that are difficult to evaluate using conventional approaches. Although multicriteria decision-making (MCDM) methods can support complex healthcare decisions

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involving multiple criteria, few studies have systematically examined the interrelationships and priorities of factors associated with nosocomial COVID-19 infection in hospital settings.

A methodological framework that integrates complementary MCDM techniques may provide a more comprehensive assessment of these risks. DEMATEL can identify the causal relationships and degree of influence among risk factors, thereby revealing which factors function as causes or effects. ANP can incorporate these interdependencies when determining the relative weights and overall importance of the factors. IPA can then compare the importance and performance of the identified factors to determine priorities for management and intervention. Integrating these three methods therefore enables the study to move beyond identifying important factors alone by simultaneously examining their causal relationships, relative importance, and practical intervention priorities.

Accordingly, this study first identified critical factors associated with hospital nosocomial infection through a literature review and then developed an integrated DEMATEL–ANP–IPA MCDM framework to prioritize these factors in a Taiwanese hospital. The findings are intended to provide hospital decision-makers with a systematic basis for strengthening infection-control strategies, improving the safety of patients and healthcare workers, and supporting risk-based resource allocation.

Literature Review

Factors Associated with Nosocomial Infection in Hospitals

Nosocomial infections occur due to several different types of factors, such as asymptomatic COVID-19 cases, delayed testing due to a lack of awareness of symptoms by healthcare workers, inadequate testing capabilities for polymerase chain reactions (PCRs) in hospitals, cross-infections due to staff movements, insufficient infection control measures, delayed treatment, close care between staff and patients, cross-infections of patients and staff, staff not wearing masks in the rest space, improper use of personal protective equipment (PPE) close contact, and other factors. According to a previous study, approximately 41% of hospital patients during the pandemic were infected and suspected to have been infected through human-to-human transmission within the hospital environment.[4] At the beginning of 2023, this study collected and categorized the literature related to the factors that lead to nosocomial COVID-19 infection in hospitals published from 2020 to 2022. The following is a literature review of the factors that lead to nosocomial infection in hospitals.

Superspreading Event

A superspreading event refers to the rapid accumulation of cases of infection by multiple transmissions over a short period.[5] During the pandemic, it was described as an estimated 80% of infections caused by approximately 10% of infected people, highlighting the importance of cluster factors in the spread of infection.[6] Patients with chronic cardiovascular disease and COVID-19 often have worse outcomes during hospitalization, and patients with respiratory symptoms that may mimic those of heart failure may have more droplets. This leads to superspreading events involving patients with high viral loads, droplet transmission, and close contact.[7] In centrally air-conditioned hospital settings, close contact with asymptomatic COVID-19 patients may result in prolonged exposure to virus-laden respiratory droplets and contaminated environments, potentially increasing viral load and the risk of severe disease and mortality among infected individuals.[8] According to previous research, the viral content is proportional to the severity of symptoms and varies with the duration of infection.[9] The high viral load in confirmed patients is correlated to the severity of other coronavirus outbreaks, and patients with high viral loads also present more severe symptoms.[10] High viral loads can also be transmitted through droplets and aerosols to patients with underlying diseases, such as chronic diseases and hypertension.[3] Moreover, children with comorbidities have also been shown to have a high viral load and an increased risk of transmission.[11] COVID-19 patients contribute to environmental contamination in hospitals through respiratory droplet transmission, with the virus being deposited on equipment such as vents through airborne transmission.[12] Whole-genome sequencing revealed that most hospital outbreaks occurred in patients who required intensive care, and the transmission routes were close contact or droplet transmission, with a mortality rate of up to 33%.[13]

Mismanagement

Mismanagement includes incomplete cleaning, asymptomatic infections, misdiagnosis, untimely screening, and improper patient triage. The primary route of transmission of the virus is through droplets from confirmed patients, which can be inhaled by nearby individuals or deposited on surfaces. Touching a contaminated surface and transferring the virus to the face has been identified as a potential route of infection.[14] It was estimated that asymptomatic infections accounted for 80.5% of total COVID-19 infections.[15] Asymptomatic carriers can spread the virus to easily susceptible individuals, even though they may not develop symptoms.[16] The chance of transmission is lower than from symptomatic patients, and some asymptomatic patients are less contagious than symptomatic patients.[17] Asymptomatic COVID-19 infection is less likely to spread than symptomatic COVID-19 infection; however, COVID-19 can result in severe disease regardless of whether it was acquired from a symptomatic or asymptomatic person, as the virus is highly transmissible.[15] Asymptomatic transmission poses a significant threat to public health, although the risk of transmission is low.[18] COVID-19 symptoms can resemble those of the common cold or influenza; furthermore, as a novel virus, COVID-19 may be misdiagnosed as other illnesses,[1] potentially preventing healthcare workers from taking immediate protective measures. Research has shown that frontline healthcare workers were infected due to staff members' lack of knowledge of the disease and inadequate protective measures.[19] Limited PCR testing capacity, delayed analysis reports, and hospital nosocomial infections can allow infections to spread while awaiting results.[20] With the increase in asymptomatic transmission and confidence in negative PCR results, the definition of suspected cases and diagnostic criteria need to be reevaluated. If a positive patient enters an area designated non-COVID-19, there is a risk of a "superspreading event" where transmission becomes difficult to track and control.[21]

Lack of Hospital Protective Equipment

The risk of COVID-19 transmission is most likely to result from prolonged unprotected contact with a symptomatic person.[22] The virus can be transmitted by patients and healthcare workers who wear masks for extended periods without changing them and by family members or staff who wear PPE incorrectly or remove masks while eating and drinking.[17] Previous studies have shown that inadequate PPE use, non-mask interactions, and failure to maintain social distancing among paediatric intensive care unit staff increased COVID-19 infection rates.[23] PPE shortages have been directly linked to nosocomial transmission, and an insufficient supply of PPE is strongly correlated with an increased risk of infection in healthcare workers,[24] which can lead to nosocomial infections.[3] Crowd density in enclosed spaces positively correlates with infection rates, with poorly designed spaces increasing patient length of stay.[25] Furthermore, compared with handheld contact ear thermometers, non-contact infrared thermographic cameras have lower sensitivity and are susceptible to environmental temperature fluctuations, potentially leading to missed detection of cases presenting with fever upon admission.[26]

Underlying Disease Risk

Hospitalized patients with COVID-19 may experience higher mortality because the severity of illness requiring hospitalization is often compounded by underlying diseases, which reduce physiological resilience and increase the risk of disease progression and death. During the pandemic, interruptions in routine care may further exacerbate preexisting conditions, delay necessary treatment, and increase vulnerability to healthcare-associated cluster infections, highlighting the need to prioritize these patients in hospital risk assessment, infection prevention, and resource allocation strategies.[27,28] Immunocompromised patients are also at increased risk of severe or complex influenza-related illness, which may be more difficult to prevent with existing vaccines and antiviral therapies. These patients are more susceptible to prolonged infections, during which viral evolution may result in mutations associated with antibody evasion and potentially increase disease burden.[29] Studies of immune-related diseases have also shown that children with cancer are particularly susceptible to infection.[30] Similarly, patients hospitalized with human immunodeficiency virus (HIV)-related COVID-19 tend to be younger than the general COVID-19 population, while multiple complex comorbidities and frequent contact with healthcare settings may increase their risk of nosocomial infection and severe COVID-19.[31]

Strategies to Prevent Nosocomial Infection in Hospitals

Monitoring and controlling nosocomial infection would be an essential daily task for hospitals, and each institution should implement effective programs according to the specific situation.[3] In particular, the high rate of pneumonia infections among COVID-19 patients in the ICU requires planning for diagnosis and measures related to infection control and prevention.[32] The strategies for preventing hospital nosocomial infection can be categorized into four aspects, discussed below.

Preparation and Education

Before infected patients enter the hospital, healthcare workers should receive infection-prevention education and practical training in appropriate infection-control procedures, while maintaining precautionary vigilance to facilitate early identification of potential transmission risks and timely implementation of preventive measures.[33] Additionally, education, training, surveillance, feedback, and organizational support are necessary to encourage behavioural change.[34] In a study of hospital employees conducted in Japan, high compliance with hand hygiene was found to reduce the risk of infection in everyday settings and prevent hospital-based nosocomial infection.[35] COVID-19 meetings were organized in response to the pandemic to update the status of the epidemic, assess infection control measures, consolidate information, and strengthen interdepartmental collaboration.[33] Accurate dissemination and active communication are also essential for correctly using PPE.[36] In PPE practices, using high-quality masks effectively reduces the risk of indoor infection caused by aerosols.[37] PPE is a crucial, effective measure for blocking the virus.[38] Regularly checking the PPE storage method and the safety equipment inventory is necessary since medical staff need sufficient protective equipment.[33] More training programs on PPE use are needed to prepare healthcare workers for safe PPE practices[39] and to improve compliance with PPE use and hand hygiene practices.[40]

Monitoring, Patient Flow Scheduling and Hospital Zoning

Identifying and isolating asymptomatic patients is crucial for nosocomial infection prevention. This includes close surveillance and screening of individuals in close contact with confirmed or suspected cases and centralized isolation of COVID-19 patients for medical observation and related testing whenever possible.[41] A simple PCR screening strategy is sufficient to detect employees with confirmed COVID-19.[42] Hospital areas should be divided. Inpatient wards for COVID-19 patients and suspected patients should be spaced separately, with negative pressure isolation devices.[43] Careful placement of COVID-19 patients, management of classified diseases, and triage strategies also help reduce contact between susceptible and infected patients. Suspected cases are screened[3] and evaluated through a preadmission travel history, occupation, contact history, and cluster (TOCC) (contact and occupation) mechanism[32] to reduce the risk of viral entry into the hospital. Alert records/logs should be continuously maintained, allowing caregivers and residents to review and follow instructions and minimize the chance of missing a COVID-19 diagnosis.[44] To reduce the exposure of patients and staff to the virus, suspected patients were sent to an isolation ward for testing.

General Logistics Work

General logistics work includes managing and disinfecting medical waste and environmental cleaning. To prevent infection, double bagging and disinfection are required for medical waste from patients with suspected or confirmed COVID-19.[45] Bedspreads, pillowcases, and other items used in isolation wards should be disinfected before washing. Cleaning and transportation personnel should also wear PPE and implement hand hygiene and disinfection measures to avoid infection.[46] Environmental surfaces in hospitals should be cleaned at least twice a day and 4 times a day in isolation wards. UV disinfection can also be performed in closed environments to enhance disinfection of patient rooms where the virus may be widely distributed.[47] Effectively protecting medical staff from COVID-19 infection begins with understanding the risk factors. Identifying these factors helps guide successful strategies to protect healthcare workers and prevent nosocomial infections in hospitals.[42]

Methods

This study proposed an MCDM model to solve this problem of evaluating hospital nosocomial COVID-19 infection control. An integrated MCDM Model which combined DEMATEL, ANP and IPA methods was conducted. The

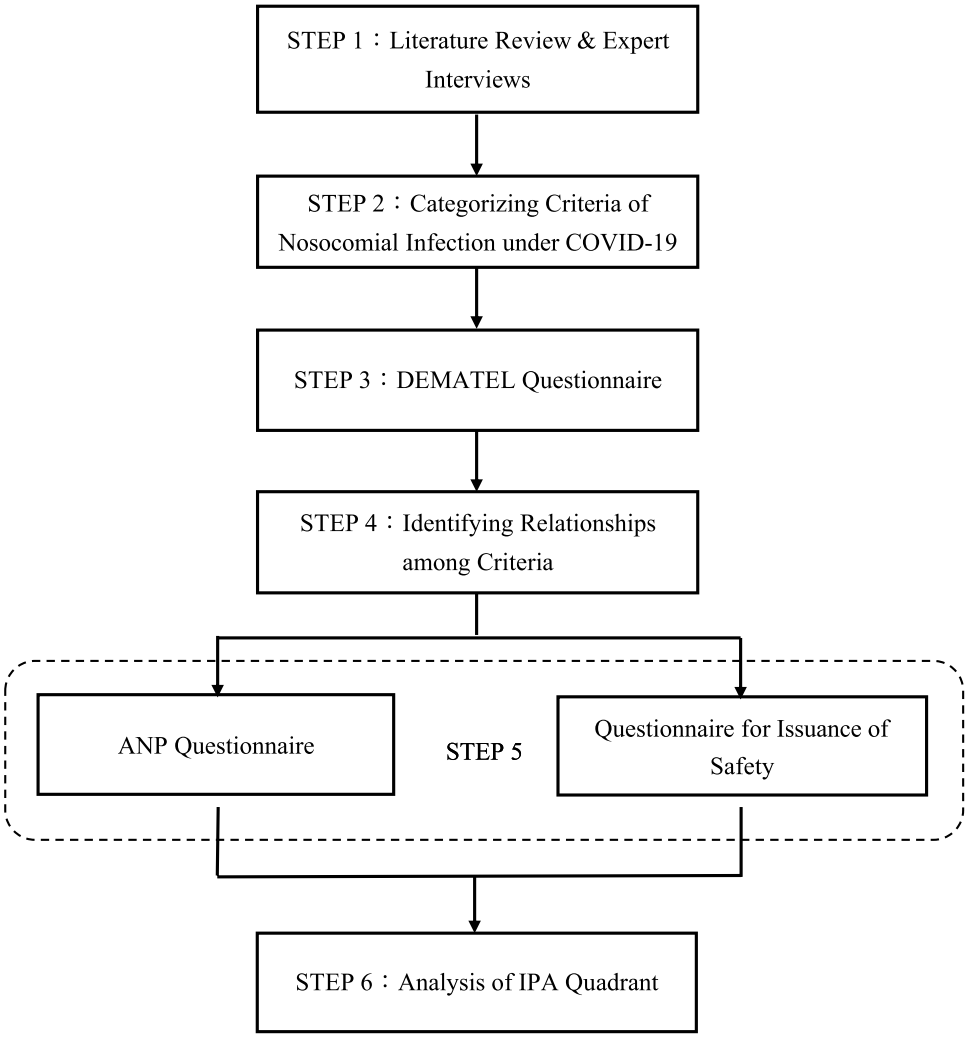


Figure 1 Research flow chart.

DEMATEL helped to investigate the causal relationships among various elements; the ANP was used to assess possible factors contributing to nosocomial infection in hospitals, with experts scoring based on the safety factor of infection control prevention in individual hospitals; IPA was employed to analyze the risk-factor prioritization of hospital investment in nosocomial infection control, providing suggestions to hospital decision-makers to improve the safety of patients and medical staff. Experts from five hospitals in Taiwan, which had frequent contact with confirmed patients, were considered for this study. A hospital in Taipei City was selected as the case example of risk-factor prioritization. The MCDM model can be divided into six steps in Figure 1.

Step 1: This study conducted a literature review and expert interviews on the factors that lead to the occurrence of nosocomial infection in hospitals.

Step 2: Based on step 1, summarize the factors that lead to the occurrence of nosocomial infection in the hospital, and develop a hierarchical structure chart. A total of four attributes and 12 criteria were summarized, as shown in Figure 2.

Step 3: The DEMATEL questionnaire was scored by comparing the variables in pairs by the 5-level scores, including 0, 1, 2, 3, and 4 (from “on impact” to “very high impact”). The definitions of attributes and criteria in the DEMATEL questionnaire are detailed in Table 1. A total of 8 questionnaires were distributed and all were collected from July to September 2022, in which there were 2 public health experts, namely Professor of Public Health (16 years seniority) and Associate Professor of Public Health (24 years seniority), 3 infection control specialists, namely nurse (25 years seniority), infection controller (22 years seniority), and infection department director (7 years seniority), and 3 medical

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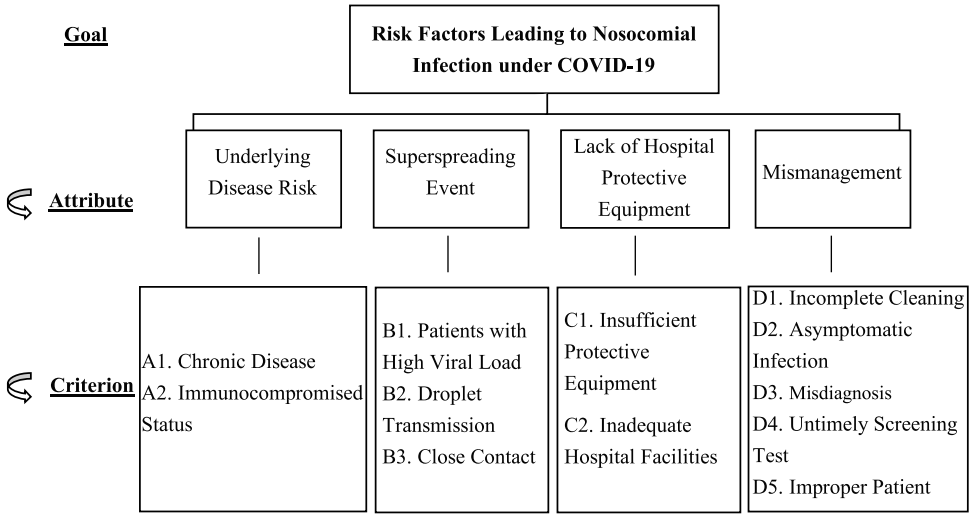


Figure 2 Factors leading to nosocomial infection under COVID-19.

management experts, including 2 medical administrative directors (more than 15 years of experience) and a deputy medical management directors/general managers (30 years of experience). This study does not involve human experimentations or participants' data. All the expert panel scorers voluntarily joined this study with consent and were only asked to identify the interactions among the attributes and criteria. As there is no human experimentation involved in this study, the study was conducted according to the guidelines of the Declaration of Helsinki and Ethical Committee protocol was not required. After aggregating the results of the expert's answers, direct/relation relationship matrix is generated. From the direct/indirect relationship matrix, the D and R values are found. The D value represents the degree to which the attribute directly or indirectly affects other attributes, while the R value represents the degree to which the attribute is directly or indirectly affected by other attributes.

Table 1 Definitions of Risk Factors

Attribute	Criteria	The Definition of Criteria
Underlying Disease Risk (UDR)	Chronic Disease (A1)	A chronic disease is a condition that persists for one year or longer, requires ongoing medical care, or limits activities of daily living. Patients with chronic diseases are at increased risk of severe COVID-19 and cluster infections.
	Immunocompromised Status (A2)	Immunocompromised patients are at increased risk of severe or complicated infections and prolonged viral infections. Prolonged infection may facilitate viral evolution, potentially contributing to immune evasion and an increased disease burden.
Superspreading Event (SE)	Patients with High Viral Load (B1)	A high viral load in infected patients may be associated with increased disease severity and a greater potential for viral transmission, thereby increasing the risk of nosocomial infection.
	Droplet Transmission (B2)	Transmission of the virus through respiratory droplets released by infected individuals, particularly during close interactions or activities that generate respiratory secretions, may increase the risk of nosocomial infection.
	Close Contact (B3)	Close contact refers to direct or prolonged interaction with an infected person during the infectious period, particularly when healthcare workers provide close-range care to patients. Such interactions may increase exposure to respiratory droplets and other infectious secretions and consequently increase the risk of nosocomial transmission.

(*Continued*)

Table 1 (Continued).

Attribute	Criteria	The Definition of Criteria
Lack of Hospital Protective Equipment (LHPE)	Insufficient Protective Equipment (C1)	PPE shortages and insufficient supply may lead to re-broadcasting of the virus in hospitals. Patients and medical staff wear masks for a long time without changing them, and some family members of patients and even staff members wear personal protective equipment incorrectly, which may cause nosocomial infection.
	Inadequate Hospital Facilities (C2)	Inadequate hospital facilities, including insufficient isolation capacity, poor ventilation, limited spatial separation, and inadequate environmental conditions, may increase the risk of healthcare-associated transmission. Overcrowding and insufficient separation of patients with different infection risks may further complicate infection prevention and control.
Mismanagement (M)	Incomplete Cleaning (D1)	Incomplete cleaning and disinfection of contaminated surfaces and objects may facilitate indirect transmission of the virus in healthcare settings. Failure to perform appropriate hand hygiene after contact with patients or contaminated surfaces may further increase the risk of nosocomial infection.
	Asymptomatic Infection (D2)	Asymptomatic infection may be difficult to identify through symptom-based screening because infected individuals may not exhibit recognizable clinical symptoms. Undetected infections may therefore delay isolation and infection-control measures and increase the potential for nosocomial transmission.
	Misdiagnosis (D3)	Overlapping clinical manifestations of COVID-19 with other respiratory infections may complicate early diagnosis, particularly when patients present with nonspecific symptoms. Misdiagnosis or delayed diagnosis may postpone appropriate isolation and infection-control measures, increasing the risk of nosocomial transmission.
	Untimely Screening Test (D4)	Untimely Screening or delayed availability of diagnostic test results may postpone the identification and isolation of infected patients. Such delays may increase opportunities for exposure and nosocomial transmission while patients remain in shared hospital environments.
	Improper Patient Triage (D5)	Inappropriate separation and placement of suspected, confirmed, and non-infected patients may increase opportunities for nosocomial transmission. Inadequate patient triage may also complicate infection-control measures, contact tracing, and timely isolation.

Step 4: Calculate the prominence (D+R) and relation (D-R) from the data results of the DEMATEL questionnaire. Prominence represents the impact of this variable and the extent to which it is affected and can tell which structural project is the core project. Relation can understand the degree of variables, if the value is positive, it represents the cause, if it is a negative table, it represents the bias as the result. Map the network relationship map to confirm the interaction between attributes and criteria. Plotting causality simplifies the complex relationships between variables and shows how each variable affects other variables. When drawing a causal diagram, prominence (D+R) is used as the horizontal axis and relation (D-R) is used as the vertical axis to form a coordinate figure.

Step 5: ANP questionnaire design and distribution, and nosocomial infection control safety factor questionnaire distribution. The evaluation scale is scored from 1 to 9, the higher the score means the higher its importance, confirm the relative importance of the variables in each level, and establish a pairwise comparison matrix to calculate its eigenvalue and eigen value. The ANP questionnaire is deferred in pairwise comparison, and experts must be required to fill in the consistency when comparing pairs. The CR (Consistency Ratio) value needs to be calculated together with the weight calculation to ensure consistency. The CR value is determined with CI/RI (<0.1) as the best. CI is consistency index, and RI is random index. The ANP questionnaire was designed based on the dependence among attributes and criteria confirmed by the statistical results of DEMATEL and was distributed at the same time with the safety assessment

questionnaire of hospital nosocomial infection. A total of 8 expert questionnaires were distributed to the ANP questionnaire, which was the same as the expert target of the DEMATEL questionnaire. However, two of the experts did not respond due to busy clinical work (Medical Administration Officer and Director of Infectious Diseases), so a total of 6 ANP questionnaires were collected. The safety assessment questionnaire was issued to score the risk factors of nosocomial COVID-19 infection control according to the performance of the case hospital from the beginning of pandemic to September 2022. All respondents were requested to score from 1 to 10 points, the higher the score means the better performances of intervention priorities of risk factors. This study proposed an MCDM model to solve this problem of evaluating hospital nosocomial COVID-19 infection control. A total of 3 copies of the safety assessment questionnaire for hospital nosocomial infection were collected, which including 2 infection control experts and 1 medical administrative director who all work in the case hospital during the pandemic.

Step 6: IPA can be a comparative analysis of attributes in quadrants, dividing attributes into four quadrants by measuring the importance and degree of expression of each attribute, providing valuable information for decision-makers to evaluate. The X-axis in the quadrant represents the criteria weight value of ANP, and the Y-axis represents the score of hospital safety factor.

Results

Dematel

Table 2 and Table 3 show the impact and degree of influence between the four attributes and the 12 criteria. The threshold for significance was set as the third quartile (Q3). The threshold value for the attribute is 3.584, and for the criterion is 0.708. If the value exceeds these thresholds, it indicates a significant impact relationship. The network relationship map of attributes and criteria is drawn (Figure 3a and b).

ANP and the Hospital Safety Scores

From Table 4, all CR values of less than 0.1 meets the consistency test requirements in ANP. According to the ANP weights, the most critical attribute leading to hospital nosocomial infection was "Lack of hospital protective equipment (LHPE)" (40.21%). The most crucial criterion for leading to hospital nosocomial infection was "Close contact (B3)" (21.28%). "Misdiagnosis (D3)", "Untimely screening test (D4)", and "Improper patient triage (D5)" were not weighted in the ANP results, and in the DEMATEL analysis, those factors were in the cause group. This means that these factors might not have the direct only indirect impact on hospital nosocomial infection of COVID-19. For the hospital safety score, all respondents were requested to score from 1 to 10 points, the higher the score means the better performances of intervention of the risk factors for case hospital. No matter from the sight of the infection control experts or medical administrative expert, the case hospital received its highest safety score in "Insufficient protective equipment (C1)"; this indicates a consensus among experts that the hospital has implemented effective interventions to address the issue of inadequate protective equipment.

Table 2 The Total Impact Relationship Matrix of Attributes

	UDR	SE	LHPE	M	D	R	D+R (Rank)	D-R (Rank)
UDR	2.916	3.370	2.944	3.193	12.423	13.351	25.774 (4)	−0.928 (4)
SE	***3.584***	***3.586***	3.391	***3.665***	14.226	14.343	28.569 (1)	−0.117 (2)
LHPE	3.490	***3.758***	3.070	3.562	13.880	12.580	26.460 (3)	1.300 (1)
M	3.362	***3.628***	3.175	3.201	13.366	13.621	26.987 (2)	−0.255 (3)

Notes: The numbers in italics and bold indicate the values≥ threshold value 3.584. D is the sum of the columns of matrix *T*, and R is the sum of the rows of matrix *T*. D+R is the prominence, and D−R is the relation.

Abbreviations: UDR, Underlying Disease Risk; SE, Superspreading Event; LHPE, Lack of Hospital Protective Equipment; M, Mismanagement.

Table 3 The Total Impact Relationship Matrix of Criteria

	A1	A2	B1	B2	B3	C1	C2	D1	D2	D3	D4	D5	D	R	D+R (Rank)	D-R (Rank)
A1	0.535	0.611	0.675	0.663	0.659	0.624	0.529	0.554	0.589	0.533	0.553	0.565	7.090	8.041	15.131 (8)	−0.952 (12)
A2	0.625	0.568	0.703	0.693	0.687	0.650	0.552	0.579	0.615	0.556	0.577	0.589	7.397	8.178	15.574 (6)	−0.781 (11)
B1	***0.758***	***0.770***	***0.742***	***0.823***	***0.822***	***0.784***	0.676	***0.713***	***0.743***	0.671	0.698	0.704	8.905	8.888	17.793 (1)	0.016 (6)
B2	***0.724***	***0.738***	***0.798***	0.706	***0.789***	***0.755***	0.650	0.692	***0.721***	0.648	0.667	0.677	8.565	8.786	17.351 (3)	−0.221 (10)
B3	***0.743***	***0.755***	***0.821***	***0.810***	***0.723***	***0.772***	0.665	0.707	***0.740***	0.663	0.682	0.701	8.782	8.779	17.561 (2)	0.003 (7)
C1	0.707	***0.720***	***0.781***	***0.773***	***0.772***	0.648	0.618	0.662	0.698	0.616	0.645	0.653	8.295	8.320	16.615 (4)	−0.026 (8)
C2	0.647	0.659	***0.710***	0.705	0.702	0.658	0.511	0.611	0.639	0.567	0.597	0.604	7.610	7.155	14.765 (10)	0.455 (2)
D1	0.665	0.678	***0.733***	***0.725***	***0.724***	0.691	0.600	0.553	0.649	0.587	0.598	0.613	7.815	7.535	15.350 (7)	0.280 (4)
D2	***0.737***	***0.751***	***0.811***	***0.800***	***0.808***	***0.762***	0.656	0.676	0.644	0.652	0.676	0.684	8.658	7.933	16.591 (5)	0.725 (1)
D3	0.635	0.644	0.706	0.693	0.698	0.660	0.566	0.596	0.622	0.503	0.598	0.607	7.527	7.130	14.657 (12)	0.398 (3)
D4	0.641	0.650	***0.715***	0.705	0.704	0.661	0.569	0.599	0.644	0.582	0.528	0.607	7.604	7.395	14.999 (9)	0.210 (5)
D5	0.624	0.634	0.694	0.690	0.689	0.655	0.563	0.592	0.628	0.552	0.576	0.524	7.422	7.529	14.950 (11)	−0.107 (9)

Notes: The numbers in italics and bold indicate the values≥ threshold value 0.708. A1=Chronic disease; A2= Immunocompromised status; B1=Patients with high viral load; B2=Droplet transmission; B3=Close contact; C1=Insufficient protective equipment; C2=Inadequate hospital facilities; D1=Incomplete cleaning; D2= Asymptomatic infection; D3= Misdiagnosis; D4=Untimely screening test; D5=Improper patient triage. D is the sum of the columns of matrix *T*, and R is the sum of the rows of matrix *T*. D+R is the prominence, and D−R is the relation.

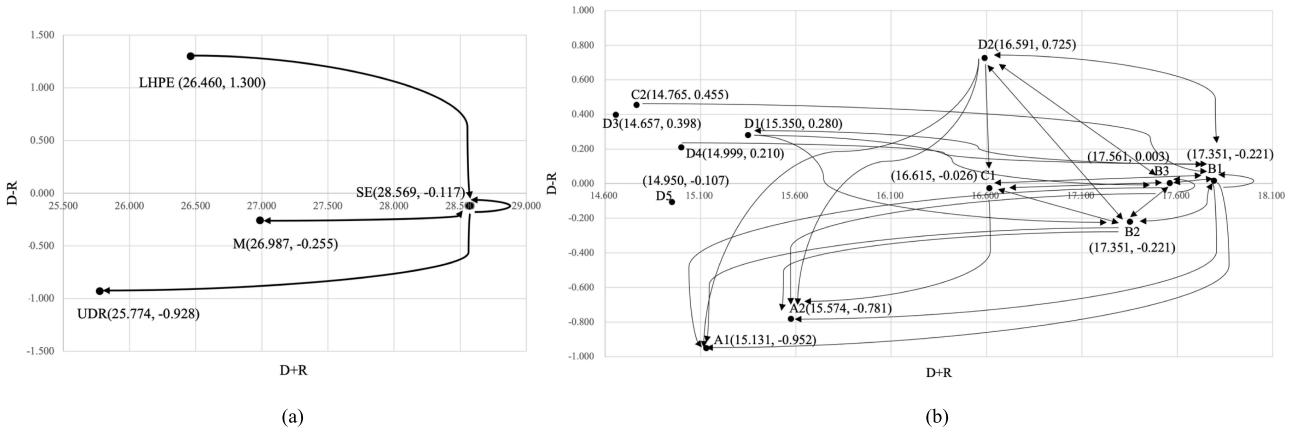


Figure 3 Influence Relationship Map of Attributes and Criteria.

Notes: (**a**) displays the influences among attributes; (**b**) displays the influences among criteria. The X axis is D+R; the Y axis is D-R. A1=Chronic disease; A2= Immunocompromised status; B1=Patients with high viral load; B2=Droplet transmission; B3=Close contact; C1=Insufficient protective equipment; C2=Inadequate hospital facilities; D1=Incomplete cleaning; D2= Asymptomatic infection; D3= Misdiagnosis; D4=Untimely screening test; D5=Improper patient triage. D is the sum of the columns of matrix *T*, and R is the sum of the rows of matrix *T*. D+R is the prominence, and D−R is the relation. ⟶ represents the unidirectional influence; ⟷ represents the two-way influence.

Abbreviations: UDR, Underlying Disease Risk; SE, Superspreading Event; LHPE, Lack of Hospital Protective Equipment; M, Mismanagement.

IPA

Figure 4 displays the IPA matrix divided by the mean (0.08) of the ANP weights (X-axis) and the mean (7.78) of the overall hospital safety factor score (Y-axis). The criteria of "Immunocompromised status (A2)", "Droplet transmission (B2)", "Close contact (B3)", and "Insufficient protective equipment (C1)" fall into the first quadrant. "Patients with high viral load (B1)", and "Asymptomatic infection (D2)" are in the second quadrant. "Incomplete cleaning (D1)", "Misdiagnosis (D3)", "Untimely screening test (D4)", and "Improper patient triage (D5)" are in the third quadrant. "Chronic disease (A1)" and "Inadequate hospital facilities (C2)" fall into the fourth quadrant.

Table 4 Weights and Hospital Safety Scores

Attribute	Criterion	Weight (%)	Rank	CR (<0.1)	Mean of Hospital Safety Scores (Standard Deviation)		
					Infection Control Experts (n=2)	Medical Administrative Expert (n=1)	Overall (n=3)
UDR		23.59	3	0.097			
	A1	5.89	7	0.059	7.50 (0.71)	9.00	8.00 (1.00)
	A2	12.16	6		7.50 (0.71)	9.00	8.00 (1.00)
SE		10.60	4				
	B1	13.23	4	0.000	7.50 (0.71)	8.00	7.67 (0.58)
	B2	12.41	5		8.00 (0.00)	8.00	8.00 (0.00)
	B3	21.28	1		8.00 (0.00)	9.00	8.33 (0.58)
LHPE		40.21	1				
	C1	16.55	2	0.049	8.00 (1.41)	10.00	8.67 (1.53)
	C2	0.00	9		8.00 (1.41)	9.00	8.33 (1.15)

(*Continued*)

Table 4 (Continued).

Attribute	Criterion	Weight (%)	Rank	CR (<0.1)	Mean of Hospital Safety Scores (Standard Deviation)		
					Infection Control Experts (n=2)	Medical Administrative Expert (n=1)	Overall (n=3)
M		25.60	2				
	D1	2.82	8	0.057	7.00 (0.00)	7.00	7.00 (0.00)
	D2	15.66	3		7.00 (0.00)	7.00	7.00 (0.00)
	D3	0.00	9		6.50 (0.71)	8.00	7.00 (1.00)
	D4	0.00	9		7.00 (0.00)	9.00	7.67 (1.15)
	D5	0.00	9		7.00 (0.00)	9.00	7.67 (1.15)

Note: A1=Chronic Disease; A2= Immunocompromised Status; B1=Patients with High Viral Load; B2=Droplet Transmission; B3=Close Contact; C1=Insufficient Protective Equipment; C2=Inadequate Hospital Facilities; D1=Incomplete Cleaning; D2= Asymptomatic Infection; D3= Misdiagnosis; D4=Untimely Screening Test; D5=Improper Patient Triage. n represents the number of experts. The numbers in parentheses represent the standard deviation.
Abbreviations: UDR, Underlying Disease Risk; SE, Superspreading Event; LHPE, Lack of Hospital Protective Equipment; M, Mismanagement.

Figure 5 uses only the safety scores from the infection control experts as performance scores. The IPA matrix divided by the mean (0.08) of the ANP weights (X-axis) and the mean (7.42) of the overall hospital safety factor score (Y-axis). "Immunocompromised status (A2)", "Patients with high viral load (B1)", "Droplet transmission (B2)", "Close contact (B3)", and "Insufficient protective equipment (C1)" are in the first quadrant. Only "Asymptomatic infection (D2)" is in

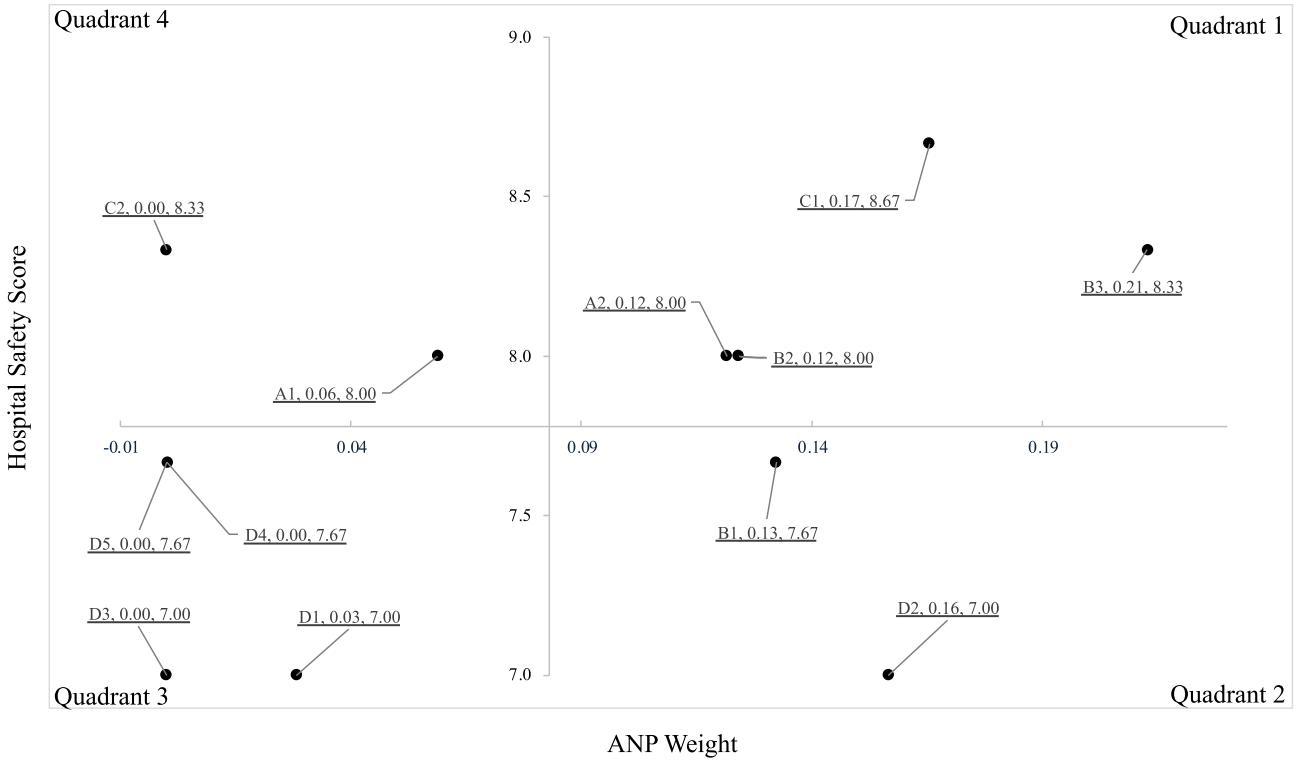


Figure 4 The IPA matrix graph – Overall safety score (n=3).
Notes: The X-axis represents the ANP weights, and the Y-axis represents the hospital safety score. A1=Chronic disease; A2= Immunocompromised status; B1=Patients with high viral load; B2=Droplet transmission; B3=Close contact; C1=Insufficient protective equipment; C2=Inadequate hospital facilities; D1=Incomplete cleaning; D2= Asymptomatic infection; D3= Misdiagnosis; D4=Untimely screening test; D5=Improper patient triage.

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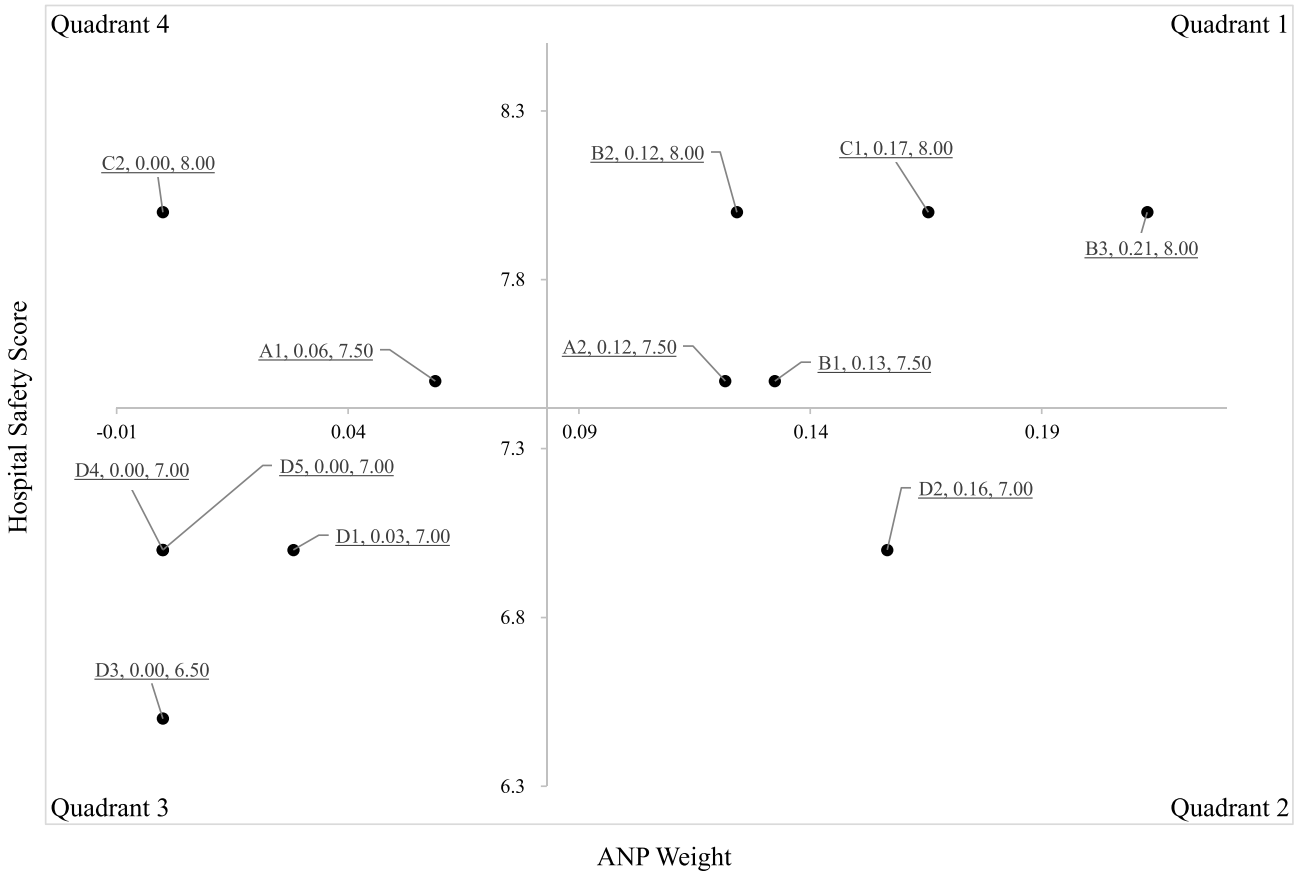


Figure 5 The IPA matrix graph – Infection control experts' safety score (n=2).
Notes: The X-axis represents the ANP weights, and the Y-axis represents the hospital safety score. A1=Chronic disease; A2= Immunocompromised status; B1=Patients with high viral load; B2=Droplet transmission; B3=Close contact; C1=Insufficient protective equipment; C2=Inadequate hospital facilities; D1=Incomplete cleaning; D2= Asymptomatic infection; D3= Misdiagnosis; D4=Untimely screening test; D5=Improper patient triage.

the second quadrant. "Incomplete cleaning (D1)", "Misdiagnosis (D3)", "Untimely screening test (D4)", and "Improper patient triage (D5)" are in the third quadrant. "Chronic disease (A1)" and "Inadequate hospital facilities (C2)" are in the fourth quadrant.

Figure 6 uses only the safety scores from the medical administrative expert as performance scores. The IPA matrix divided by the mean (0.08) of the ANP weights (X-axis) and the mean (8.50) of the overall hospital safety factor score (Y-axis). "Immunocompromised status (A2)", "Close contact (B3)", and "Insufficient protective equipment (C1)" are in the first quadrant. "Patients with high viral load (B1)", "Droplet transmission (B2)", and "Asymptomatic infection (D2)" are in the second quadrant. "Misdiagnosis (D3)" and "Incomplete cleaning (D1)" are in the third quadrant. "Chronic disease (A1)", "Inadequate hospital facilities (C2)", "Untimely screening test (D4)", and "Improper patient triage (D5)" are in the fourth quadrant.

Discussion

Regarding the four attributes, "Lack of Hospital Protective Equipment (LHPE)" received the highest score in the DEMATEL relation (D-R), indicating it is a significant contributing risk factors for nosocomial COVID-19 infection. This attribute significantly affects other attributes and holds the top rank in ANP weight, which experts consider to be the most crucial attribute in the hospital for nosocomial COVID-19 infection. As in previous studies, "Lack of Hospital Protective Equipment (LHPE)" was highly correlated with nosocomial infection.[2,48] During the COVID-19 epidemic, hospitals not only treated confirmed COVID-19 patients but continued to manage preexisting patients and related public health tasks. Even if hospitals implement measures such as screening, disinfection, or triage for COVID-19, a lack of protective equipment may promote the

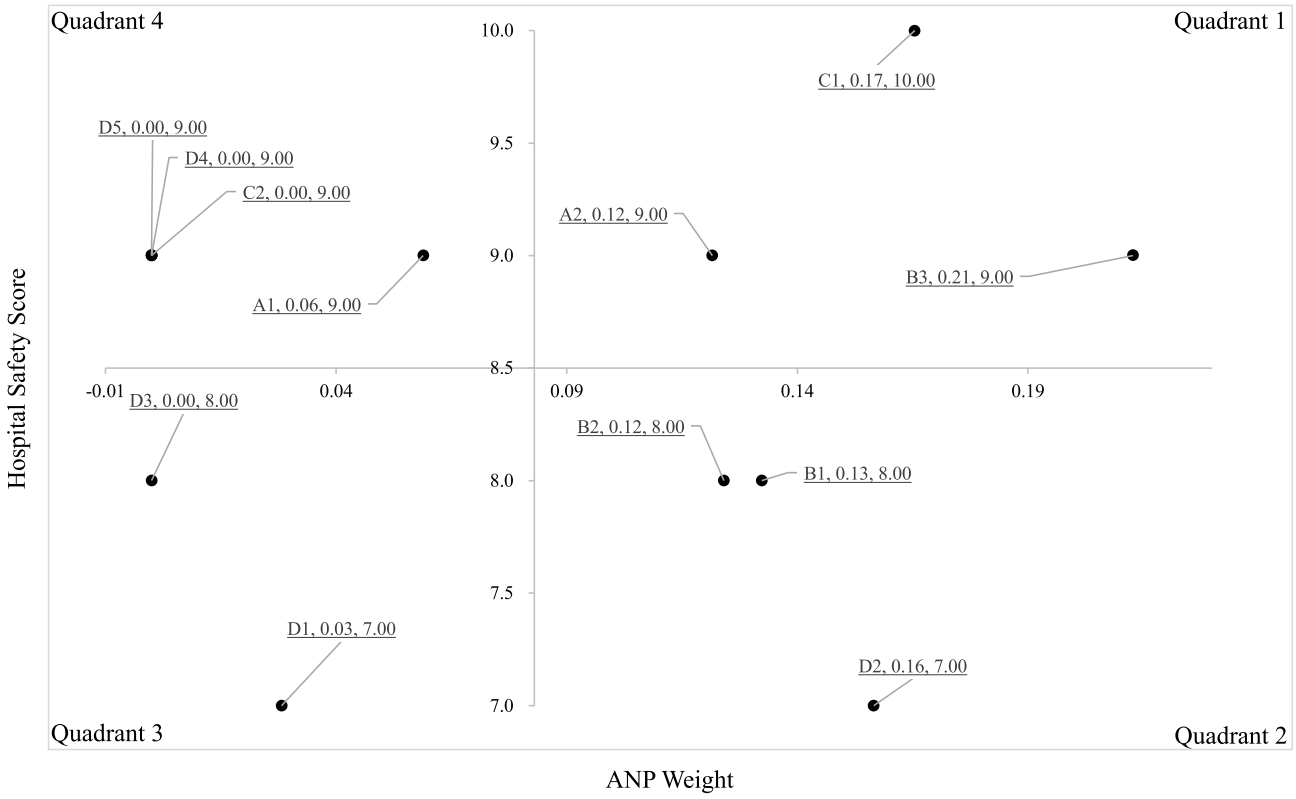


Figure 6 The IPA matrix graph – Medical administrative expert's safety score (n=1).
Notes: The X-axis represents the ANP weights, and the Y-axis represents the hospital safety score. A1=Chronic disease; A2= Immunocompromised status; B1=Patients with high viral load; B2=Droplet transmission; B3=Close contact; C1=Insufficient protective equipment; C2=Inadequate hospital facilities; D1=Incomplete cleaning; D2= Asymptomatic infection; D3= Misdiagnosis; D4=Untimely screening test; D5=Improper patient triage.

virus's spread through droplet transmission and close contact, limiting the effectiveness of these measures. Meanwhile, "Superspreading Event (SE)" ranks first in the DEMATEL prominence analysis (D+R), indicating that it was perceived by experts the most significant relationships among the other risk factors of nosocomial infections in hospitals. Although its ANP weight is ranked last, hospitals may still prioritize further prevention improvements against this attribute; this is because, according to the results of DEMATEL, "Superspreading Event (SE)" might not directly impact hospital nosocomial infection, but it could have affect by influencing other factors indirectly. "Superspreading Event (SE)" refers to the rapid accumulation of multiple infection cases in multiple transmission chains within a short period, so the occurrence of "Superspreading Event (SE)" in the hospital may overwhelm hospital capacity and disrupt hospital operations. Previous studies have shown that an estimated 80% of infections are caused by 10% of infected people,[6] demonstrating the importance of cluster factors for infections and superspreading events.

In the criteria section, "Patients with high viral load (B1)" is ranked the highest DEMATEL prominence (D+R), highlighting its crucial role in nosocomial infection in hospitals. Representative hospitals may also prioritize further preventive improvement for this criterion, and in the DEMATEL relation, it is positive, which affects the negative attribute. Patients with a high viral load are more likely to transmit the virus, especially during close contact, and are associated with more severe disease outcomes during coronavirus outbreaks.[10] Effective screening and isolation should also be carried out during close contact to prevent the spread of the virus. In previous studies, patients and doctors who wore effective protective gear and were infected by close contact were shown to have milder symptoms of COVID-19[13] and effective protection against people diagnosed with high viral loads. Also, "Close contact (B3)" is ranked second in DEMATEL prominence (D+R), indicating that it is second in importance in causing hospital nosocomial infection. Representative hospitals may also prioritize further prevention improvements against this attribute, and this criterion is also ranked first in ANP weighting. Previous studies have confirmed that "Close contact (B3)" is strongly correlated with

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the spread of the virus.[9,13] "Patients with high viral load (B1)" and "Close contact (B3)" are criteria under the "Superspreading Event (SE)" attribute.

In summary, "Lack of hospital protective equipment (LHPE)" and "Superspreading Event (SE)" would be critical attributes of nosocomial infection. The critical risk factors would be "Patients with high viral load (B1)" and "Close contact (B3)" to effectively prevent nosocomial infection.

Furthermore, IPA analyzed the importances and performances of risk factors of the case hospital in preventing nosocomial infections during the COVID-19 pandemic. The risk factors in the first quadrant indicates areas where experts perceived that the case hospital had well prepared and should continue to maintain the efforts. Overall, the risk factor of "Immunocompromised status (A2)", "Close contact (B3)", and "Insufficient protective equipment (C1)" in the case hospital was considered good preparation to prevent nosocomial infections during the COVID-19 pandemic by all experts. This is due to reduced non-emergency hospitalizations during the epidemic and the introduction of telemedicine services, which minimizes the risk of hospital visits. Comparing the different experts, the infection control experts also considered the risk factors of "Patients with high viral load (B1)" and "Droplet transmission (B2)" in the first quadrant; "Patients with high viral load (B1)" and "Droplet transmission (B2)" are the reasons why "Superspreading Event (SE)" occur. "Superspreading Event (SE)" are also considered one of the most important key attribute metrics for the occurrence of hospital nosocomial infection in the DEMATEL analysis. These results revealed the case hospital is fully prepared for the "Superspreading Event (SE)".

The guidelines issued by Taiwan government for medical institutions in response to COVID-19 infection control measures have recommended that hospitals regularly disinfect, maintain social distancing, and establish relevant management measures. For example, inventory management of protective equipment and competent health authorities should regularly provide PPE for medical institutions. In addition to complying with the requirements of Taiwan government, the case hospitals should also strengthen the in-hospital epidemic prevention requirements during the outbreak, such as banning staff from visiting non-work departments and closing floors when staff are infected. Strengthening cleaning and disinfection, such as disinfecting once daily for two hours, will also decrease transmission rates. To reduce the risk of exposure, individual hospitals should offer medication collection services outside the hospital or encourage patients to pick up medications at community pharmacies to reduce the risk of exposure. As a result, these measures are considered to perform well by the experts in the IPA analysis and should be maintained.

Criteria falling in the second quadrant represent areas that require priority improvement, while the third quadrant suggests areas for secondary improvement. Overall, "Patients with high viral load (B1)" and "Asymptomatic infection (D2)" of the case hospital were considered as the risk factors with high priority to be improved in developing hospital nosocomial infection. In DEMATEL-ANP analysis, "Patients with high viral load (B1)" is also considered as the critical factor in the spread of nosocomial infections and have impact to the other risk factors of hospital nosocomial infection. The published study had confirmed that asymptomatic infection patients are still capable of virus transmission, posing a threat to hospitals.[18] Most of the risk factors of the attribute "Mismanagement (M)" were allocated in the third quadrant. One of the secondary risk factors for improvement is "Incomplete cleaning (D1)", which is consistent with previous studies and increases the risk of infection without proper hand hygiene.[49] That is, the risk factors of "Mismanagement (M)" would also need to be emphasized after the case hospital can control the risk factors of "Patients with high viral load (B1)" and "Asymptomatic infection (D2)". During the middle and later stages of the pandemic, Taiwan government request all hospitalized patients and accompanying individuals to be screened upon admission. Regular screenings are also necessary after hospitalization, and hospital staff are regularly provided with masks. In response to the PPE shortage in hospitals during the COVID-19 outbreak, protective clothing and eye protection were provided in addition to masks in hospitals.

The fourth quadrant represents excessive resource investment. Overall, both of the infection control experts and the medical administrative expert believe that resources may be disproportionately allocated to hospital facilities, "Inadequate hospital facilities (C2)", and these resources could be shared with the second quadrant for prioritizing improvement. This may be because the case hospital was built more than 40 years ago, and although there are SOP procedures and response measures in place, the case hospital put lots of efforts to follow the requests from government and to let the physical infrastructure be adequate to handle the demands of modern infectious disease management. As

the previous publication noted, the spatial environment of medical buildings might be one of the leading causes of nosocomial infections,[25] and improving hospital facilities might be one of the most visible improvements. "Chronic disease (A1)" is also considered a risk factor with over efforts for the case hospital because the hospital serves COVID-19 patients and chronic conditions but also has a nursing home. During the early stage of the pandemic, the care hospital worried these patients are more vulnerable to infections and may put more extra measures to reduce their exposure risk.

Based on the results of IPA, the case hospital should give insights regarding hospital infection prevention strategies, particularly in addressing the patients with high viral load and asymptomatic infection. Although the case hospital is a 40-year-old building with an attached nursing home, redesigning dedicated wards in response to COVID-19, disease management, and TOCC mechanisms should be considered. For minor improvements, the medical waste of COVID-19 patients should be double-bagged,[47] and PPE should be prepared and used consistently[39,40] to address incomplete cleaning and ensure proper screening measures.[42] Additionally, confirmed staff or patients should be detected in real-time in response to screening, not delayed. The ward space for confirmed and suspected patients should be segregated[43] to address issues related to patient triage. To reduce the number of misdiagnosed patients and enhance safety, the outbreak information should be updated and integrated through pandemic meetings,[33] strengthening interdepartmental collaboration and improving the overall safety environment for patients and medical staff.

However, these recommendations and the priorities generated by the proposed MCDM framework should be interpreted in consideration of contextual differences. Infection-control requirements, particularly PPE use, may vary according to hospital type, healthcare infrastructure, national resources, and the stage of an epidemic. Measures appropriate during the early COVID-19 pandemic may differ from those required during later stages because of changes in vaccination coverage, clinical knowledge, antiviral treatments, surveillance capacity, and infection-control guidance. Hospitals in high-income countries may also have greater access to PPE, isolation facilities, and diagnostic resources than those in middle- or low-income settings. Besides, the number of experts participating in the IPA assessment was limited because the questionnaire involved sensitive issues related to hospital performance and infection-control management, which may have discouraged some potential participants. Therefore, the specific priorities and weights identified in this single-hospital Taiwanese case should not be directly generalized to other hospitals, countries, or epidemic stages without contextual reassessment. Nevertheless, the integrated DEMATEL–ANP–IPA framework may be adapted to other pandemics and healthcare settings by revising the criteria, expert composition, and weighting according to local epidemiological, organizational, and resource conditions. Future studies should validate the framework using larger and more diverse expert panels, multiple hospitals and countries, and empirical clinical or epidemiological data.

Conclusion

This study established the MCDM model that combines DEMATEL, ANP, and IPA. A case simulation analysis was conducted at an individual hospital to apply this model. The results showed that the risk attribute indicators, "Lack of Hospital Protective Equipment (LHPE)" and "Superspreading Event (SE)", were considered critical for preventing nosocomial COVID-19 infection by experts; the critical risk criteria indicators were "Patients with high viral load (B1)" and "Close contact (B3)". To effectively prevent nosocomial infections, hospitals should focus on improving these attributes and criteria, thereby minimizing the impact of the COVID-19 virus. In the case hospital study, "Patients with high viral load (B1)" and "Asymptomatic infection (D2)" were considered priority areas for improvement, followed by "Incomplete cleaning (D1)". Infection control triage should be enhanced through disease management and implementation of TOCC mechanisms. This will help reduce the infection risk of patients with high viral load and asymptomatic infection. Hospitals should monitor and implement key infection measures, including environment cleaning, providing sufficient PPE, proper use of PPE, isolating suspected or confirmed patients, and implementing elimination measures to reduce nosocomial infections and further improve the safety of patients and medical staff.

As the global epidemic has gradually subsided and the disease is now coexisting with the virus in a post-epidemic era, the severity and mortality rates of COVID-19 have also begun to rise. Hospitals must comply with government guidelines and develop internal policies and operational protocols. On the hospital side, this requires setting clear goals through organizational vision, policies, strategies, and short-term, medium-term, and long-term objectives. Through leadership and decision-making, as well as consensus camps or seminars within hospital teams, a shared understanding

can be created to promote the implementation of infection control measures in the hospital. Strategies such as TOCC implementation, screening mechanisms, and hand hygiene should be integrated into daily operations to reduce the risk of nosocomial infection. Besides, it had been highlighted the importance of strengthening communication, openness, and team cohesion in promoting infection prevention and control management strategies during the COVID-19 pandemic.

This study addresses an identified gap by developing an integrated MCDM model that combines DEMATEL, ANP, and IPA to identify, examine, and prioritize risk factors associated with hospital nosocomial infection during the COVID-19 pandemic. This model provides a systematic approach for examining the interrelationships among risk factors, determining their relative importance, and identifying priorities for infection-control intervention simultaneously. In practice, the findings may support hospital decision-makers in conducting context-specific risk assessments and allocating infection-control resources according to identified priorities. However, several limitations should be considered. The findings were based on expert judgment and a single-hospital case study in Taiwan; therefore, subjective assessment, hospital-specific characteristics, differences in healthcare systems and resources across countries, and contextual factors may influence the resulting priorities and weights. Accordingly, the findings should not be directly generalized to other hospitals, countries, or epidemic stages without contextual reassessment. Nevertheless, the proposed model may be adapted to other healthcare settings and emerging infectious diseases by revising the criteria, expert composition, and weighting according to the characteristics of the disease, healthcare system, and available resources. Future studies should validate the framework using larger and more diverse expert panels, multiple hospitals and countries, and empirical clinical or epidemiological data. This model may also provide a reference for health authorities and healthcare organizations when developing context-specific infection prevention policies and preparedness strategies for future respiratory infectious disease outbreaks.

Ethics Statement

There is no human experimentation involved in this study. As the study does not include any attributes of medical experiments, the study was conducted according to the guidelines of the Declaration of Helsinki and Ethical Committee protocol was not required.

Disclosure

The authors report no conflicts of interest in this work.

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